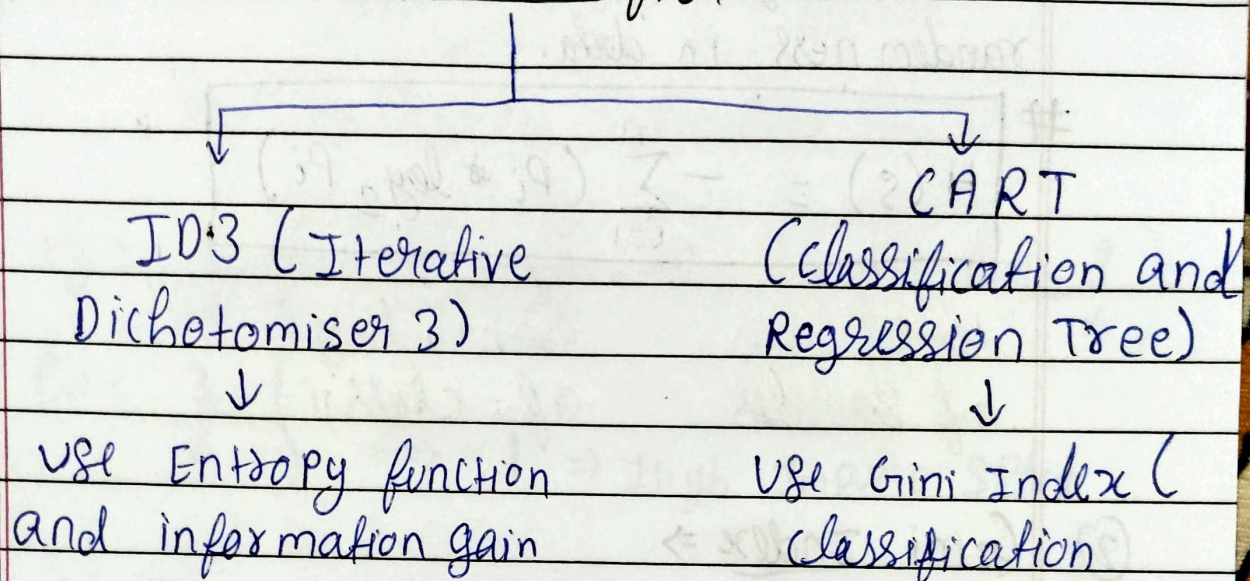


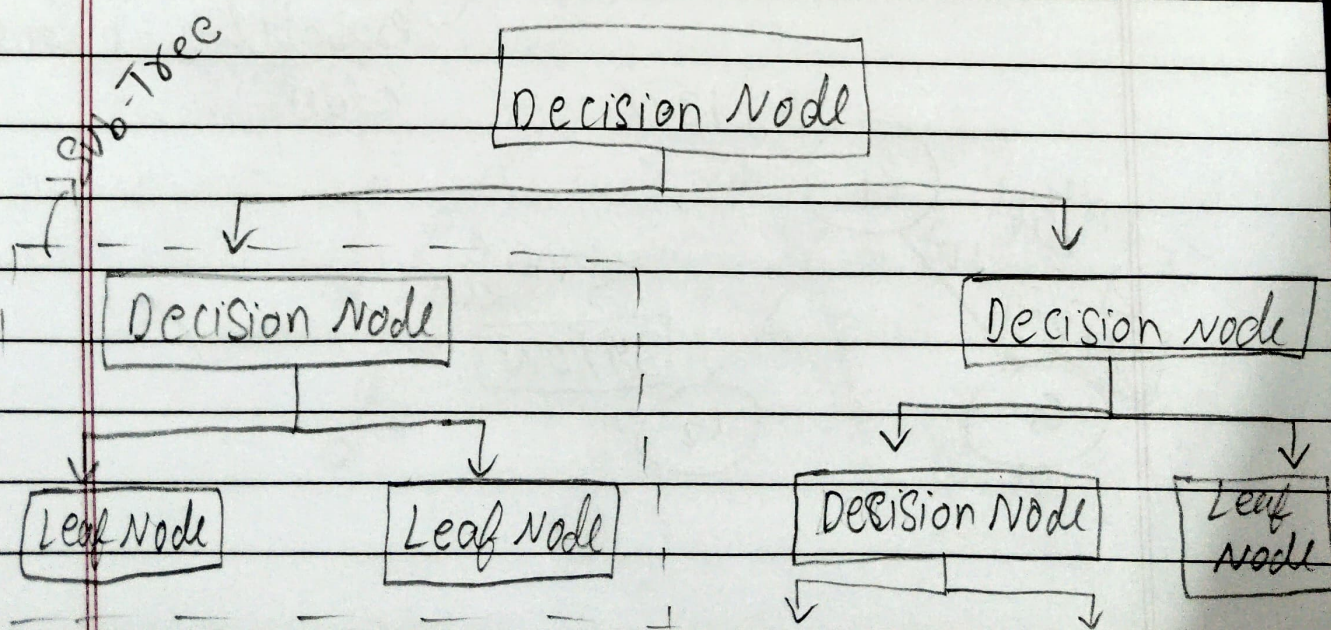
Decision Tree

① Decision Tree classifier



a) Entropy and Gini Index → Purity split

b) Information Gain → Features to select for Decision Tree construction



① Entropy $\Rightarrow$

It is a metric to measure the impurity in a given attribute. It specifies randomness in data.

#

$$H(s) = - \sum_{i=1}^n (P_i \log_2 P_i)$$

total no
of samples

probability
of class i

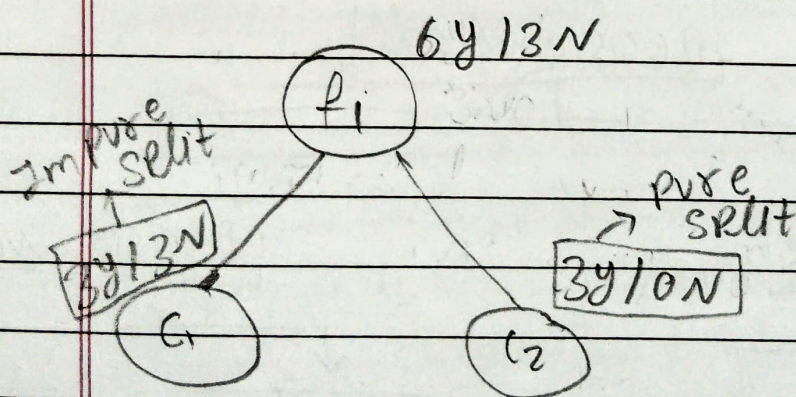
② Gini Index $\Rightarrow$

It is measure the impurity or purity used while creating the decision tree.

#

$$G.I = 1 - \sum_{i=1}^n (P_i)^2$$

$\rightarrow$ probability belonging class.



① Entropy:

$$H(C) = - \sum_{i=1}^n (P_i \log_2 P_i)$$

→ probability
of yes

→ probability
of no

$$\Rightarrow -P_y \log_2 P_y - P_N \log_2 P_N$$

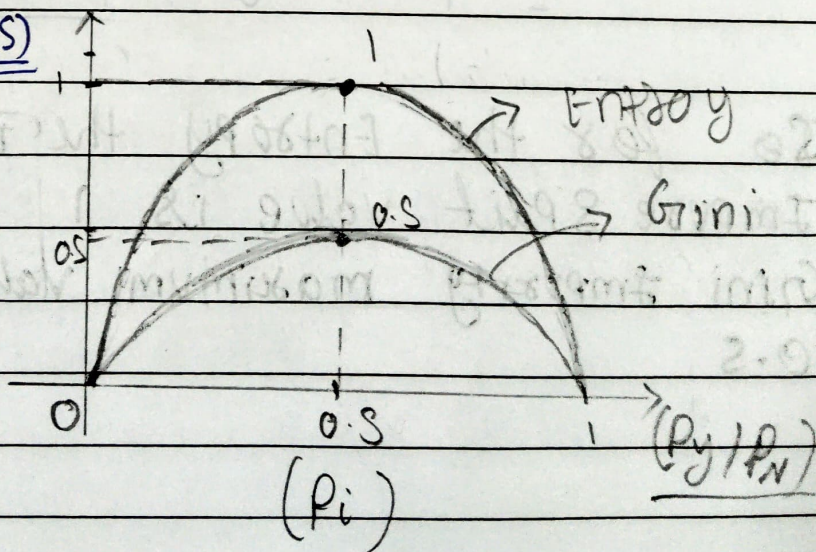
$$\left. \begin{array}{l} P_y = \frac{3}{6} \\ P_N = \frac{3}{6} \end{array} \right\} \Rightarrow -\left(\frac{3}{6} \log_2 \frac{3}{6}\right) - \left(\frac{3}{6} \log_2 \frac{3}{6}\right)$$

$$\Rightarrow 1 \Rightarrow \text{High Impure split}$$

$$H(C_2) = - \sum_{i=1}^n (P_i \log_2 P_i)$$

$$\left. \begin{array}{l} P_y = \frac{3}{3} = 1 \\ P_N = \frac{0}{3} = 0 \end{array} \right\} \Rightarrow - (1 \log_2 1) - (0 \log_2 0)$$

$$\Rightarrow 0 \Rightarrow \text{Pure split}$$

H(S)

Gini Impurity:

$$G.I = 1 - \sum_{i=1}^n (p_i)^2$$

$$G.I(c_1) = 1 - ((p_y)^2 + (p_n)^2)$$

$$= 1 - \left(\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2\right)$$

$$= 1 - \left(\frac{1}{4} + \frac{1}{4}\right)$$

$$= \frac{1}{2} = 0.5 \Rightarrow \text{High Impure Split}$$

$$G.I(c_2) = 1 - ((p_y)^2 + (p_n)^2)$$

$$= 1 - \left(\left(\frac{3}{3}\right)^2\right)$$

$$= 1 - 1 = 0 \Rightarrow \text{pure split}$$

{ So for the Entropy the maximum Impure Split value is 1 and for Gini Impurity maximum value is 0.5

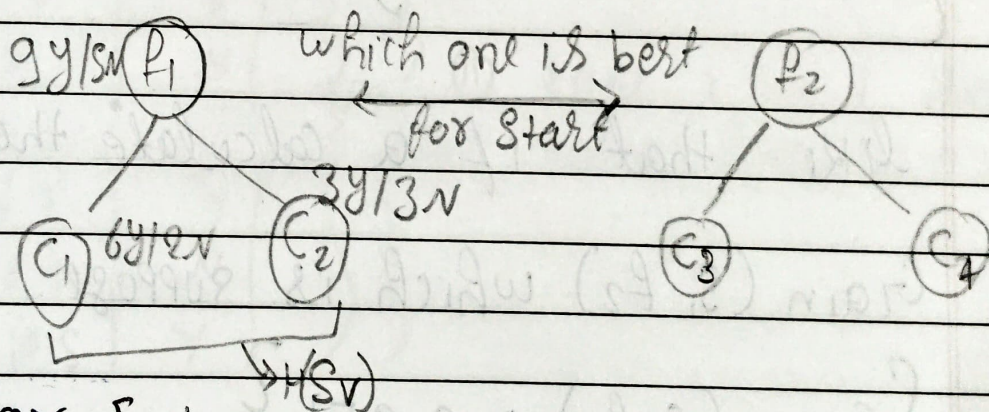
Information Gain

$$\text{Gain}(S, f_i) = H(S) - \sum_{v \in \text{val}} \frac{|S_v|}{|S|} H(S_v)$$

$\nearrow$ Entropy $\nearrow$ sub-node sample $\nearrow$ node entire sample
 $\downarrow$ v belongs to value $\downarrow$ total sample

$$H(S) = - \sum_{i=1}^n (P_i \log_2 P_i)$$

$\{f_1, f_2, f_3, o/p\}$



for f_1 :

$\rightarrow$ root node

$$H(S) = - \left(\frac{9}{14} \log_2 \frac{9}{14} \right) - \left(\frac{5}{14} \log_2 \frac{5}{14} \right)$$

$$= 0.94$$

$$H(c_1) = -\left(\frac{6}{8} \log_2 \frac{6}{8}\right) - \left(\frac{2}{8} \log_2 \frac{2}{8}\right)$$
$$= 0.81$$

$$H(c_2) = -\left(\frac{3}{6} \log_2 \frac{3}{6}\right) - \left(\frac{3}{6} \log_2 \frac{3}{6}\right)$$
$$= 1$$

$$\text{Gain}(S, f_1) = 0.94 - \left[\frac{8}{14} \times 0.81 + \frac{6}{14} \times 1 \right]$$

$$\{ \text{Gain}(S, f_1) = 0.049 \} \Rightarrow \text{for } f_1$$

like that if a calculate the
 $\text{Gain}(S, f_2)$ which is suppose.

$$\{ \text{Gain}(S, f_2) = 0.051 \}$$

$$\Rightarrow \text{So here } (\text{Gain}(S, f_2) > \text{Gain}(S, f_1))$$

than we have to start with f_2 feature.

whenever the dataset is small we have to use the Entropy in decision tree to take the decision

when dataset is large we have to use the Gini in decision tree to take the decision

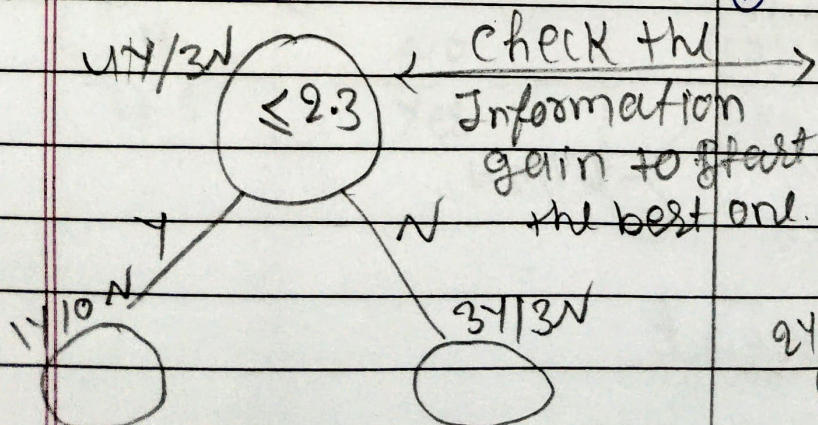
Decision tree for Numerical value.

f_1	O/P
2.3	Y
3.6	Y
4	N
5.2	N
6.7	Y
8.9	N
10.5	Y

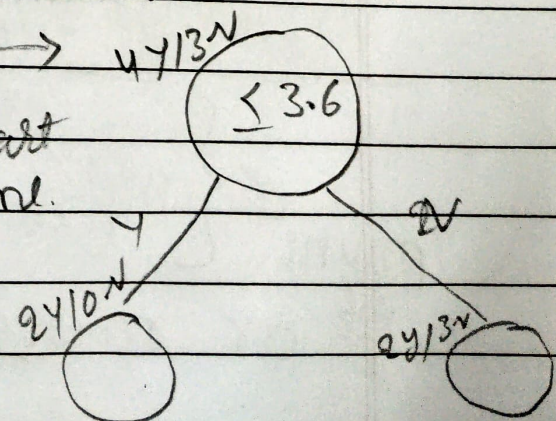
① sort the f_1 values.

② take 1 by 1 threshold value like $(2.3 \rightarrow 3.6 \rightarrow 4 \dots)$

① Threshold = 2.3



② Threshold = 3.6

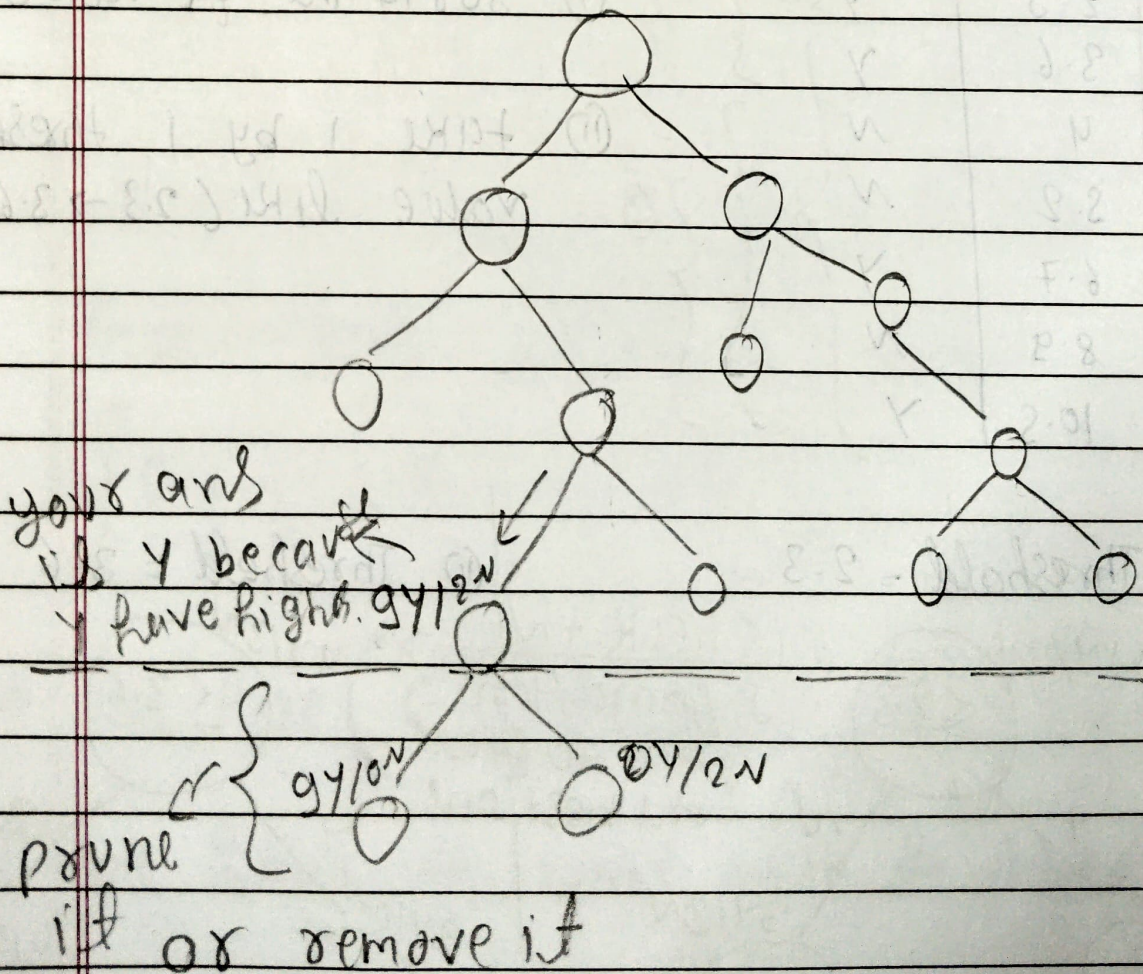


Disadvantage: if data is too large

it creates too many decision trees for the calculation which is taking time.

Post Pruning And Pre Pruning:

⇒ we use these techniques to reduce the overfitting of our model by removing the last leaf node.



- Post-Pruning $\Rightarrow$
- 1) construct a decision tree
 - 2) prune it with respect to depth
 - 3) apply for smaller dataset

Pre-Pruning $\Rightarrow$

- 1) Tune with hyperparameter when constructing the tree.
- 2) apply for large dataset.

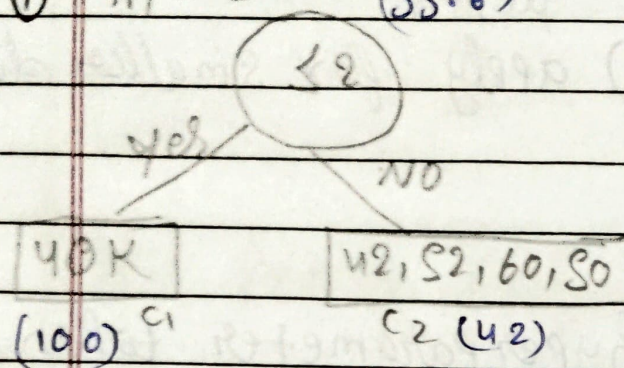
2) Decision Tree Regression.

Dataset

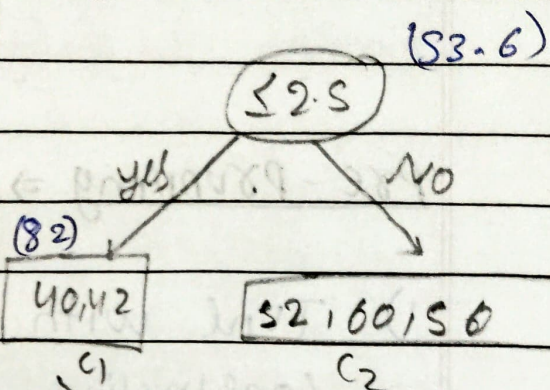
Exp	Gap	Salary
2	Yes	40K
2.5	Yes	42K
3	No	52K
4	No	60K
4.5	Yes	50K

⇒ using 2 threshold value

① $th = 2$ (S3.6)



② $th = 2.5$



← which one is selecting

we use the ~~var~~ variance Reduction for the regression problem.

$$\text{Variance} = \frac{1}{n} \sum_{i=1}^n (y - \bar{y})^2$$

$\nwarrow$ mean squared error $\searrow$ Average

{ for threshold = 2 }

$$\text{variance of root} = \frac{1}{5} \left[(40-50)^2 + (42-50)^2 + (52-50)^2 + (60-50)^2 + (50-50)^2 \right]$$

{ avg = $\bar{y}$ = 50K }

$$= \frac{1}{5} [100 + 64 + 4 + 100 + 0] = \underline{53.6}$$

$$\text{for } (C_1) \Rightarrow \frac{1}{1} (40 - 50)^2 = \underline{100}$$

$$\text{for } (C_2) = \frac{1}{4} [(42 - 50)^2 + (32 - 50)^2 + (60 - 50)^2 + 0]$$

$$= \frac{1}{4} (64 + 4 + 100) = \underline{42}$$

variance Reduction = $\text{var}(\text{root}) - \sum w_i \text{var}(C_i)$

$\Rightarrow$ variance Reduction for 1st tree!

$$\Rightarrow 53.6 - \left[\frac{1}{5} \times 100 + \frac{4}{5} \times 42 \right]$$

$$\Rightarrow 53.6 - [20 + 33.6]$$

$$\Rightarrow 53.6 - 53.6$$

variance $\Rightarrow$ 0

reduction

$\Rightarrow$ for first threshold.

For threshold = 2.5

variance (root) = 53.6

$$\begin{aligned} \text{for } (C_1) &\Rightarrow \frac{1}{2} [(40-50)^2 + (42-50)^2] \\ &= \frac{1}{2} [100 + 64] \end{aligned}$$

$$= \frac{1}{2} \times 164 = 82$$

$$\begin{aligned} \text{for } (C_2) &\Rightarrow \frac{1}{3} [(52-50)^2 + (60-50)^2 + (50-50)^2] \\ &= \frac{1}{3} [4 + 100 + 0] \end{aligned}$$

$$= \frac{104}{3} = 34.66$$

variance Reduction for second tree:

$$\Rightarrow 53.6 * \left[\frac{2}{8} \times 82 + \frac{3}{5} \times 34.66 \right]$$

$$\Rightarrow 53.6 - [32.8 + 20.796]$$

$$\Rightarrow 53.6 - 53.596 = 0.004 \Rightarrow \text{we chose the } \underline{\text{2nd tree}}$$